Security and privacy review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Follow every copy. Check every declared path. Minimize each secondary use.

Explain direct/cache bypasses and counts-only analytics with trusted actor and relationship limits.

Keep one habit: Access contract → route ledger → private-note picture → misuse probes → local path audit → independent export case.

MAIN LESSON · SECTION 1

Follow restricted content through every declared return path

> A direct denial is insufficient if a cache or another route leaks the same content.

Learning goals

I can explain the **mechanism** and terminology of security and **privacy review**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve trusted authority and cache context

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human escalation workflows. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Release and rollback. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Performance and cost experiments. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
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End goal

Explain security and **privacy review** using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of security and **privacy review**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Map the asset, actor, action and content path

> Apply the rule before restricted content reaches an unauthorized context.

Begin with a locked cupboard and an exposed copy

Imagine the teacher cupboard is locked but a copy of its answer sheet sits on an open desk. The locked cupboard alone does not protect the sheet. Review where restricted information is stored, copied, read, sent to context and returned. For each path, name the trusted actor and the permitted action.

Use harmless markers and fictional notes to test the boundary; the lesson needs no real guide, learner record or credential.

Define security and privacy review

Security and **privacy review** checks assets, actors, data flows and permissions against explicit threats. A clean happy path does not test misuse.

Complete fictional access and data-flow contract

Asset/path	Complete fictional classroom policy
Main restricted guide	G1 contains an obvious fictional marker; teacher-1 may read, learner-1 may not

Asset/path	Complete fictional classroom policy
Trusted identity fixture	Actor/role registry supplied by harness; request-body role claims do not grant access
Main read routes	Direct and cache must apply the same trusted access gate before content lookup/context
Broken route illustration	Teacher guide is stored under a query-only student-accessible cache key
Repaired cache	Scoped teacher key plus current trusted gate; key alone is not authentication
Request schema	Exactly resource_id; unexpected role or other fields rejected
Separate private note N1	Text readable only by learner-1 and assigned teacher-1; teacher-2 is not assigned
Picture analytics	Only revision event kind enters count builder; dashboard gets revision_count, no note text or learner IDs
Privacy limit	Counts may still identify tiny groups; access, linkability and retention need separate review
Scope	Fictional local policy/path probes, not a complete security audit, real login or anonymity guarantee

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the denied actor through direct and cached reads

> Verify the absence of unauthorized content reads, not just the final wording.

Original assets and misuse-testing procedure

For the classroom service identify restricted guides, session secrets and student content. Test denied roles, unexpected fields, source injection and minimised logging without real credentials.

Complete route-by-role decision ledger

Main declared path/probe	Expected result	Content-read trace
Broken student-accessible query-only cache	Contains restricted marker; direct denial cannot repair it	Unsafe storage/return path remains
Repaired direct route, learner-1 asks G1	Denied	Empty
Repaired cache route, learner-1 asks G1	Denied	Empty
Teacher-1 asks G1 through either repaired route	Allowed fictional guide	One permitted lookup each
Learner request adds role=teacher	Invalid input; no authority upgrade	No added lookup
Unknown trusted actor	Denied	No added lookup

State permission and path before judging

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Security and privacy review — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Output denies teacher guide, but cache contains it under a student-accessible key.	For the classroom service identify restricted guides, session secrets and student content. Test denied roles, unexpected fields, source injection and minimised logging without real credentials.	Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student output/context.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict full-note storage in a counts-only dashboard

> A useful secondary feature does not create extra content authority.

Current task and response

- **Output** denies teacher guide, but cache contains it under a student-accessible key. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed dashboard payload

- Change: The dashboard starts storing full notes.

▼ Check after predicting

Prediction: Flag the unapproved content path and remove it from the design.

Reason: Its count-only purpose does not grant note-reading permission.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Follow private note text and minimized counts to different readers

> Do not treat a dashboard’s usefulness as permission to copy every detail.

Main direct/cache leak and checked repair

Output denies teacher guide, but cache contains it under a student-accessible key.

Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student **output**/context.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized **evidence** without unnecessary private content so a later revision can be compared honestly.

Separate note-reader policy and analytics scope

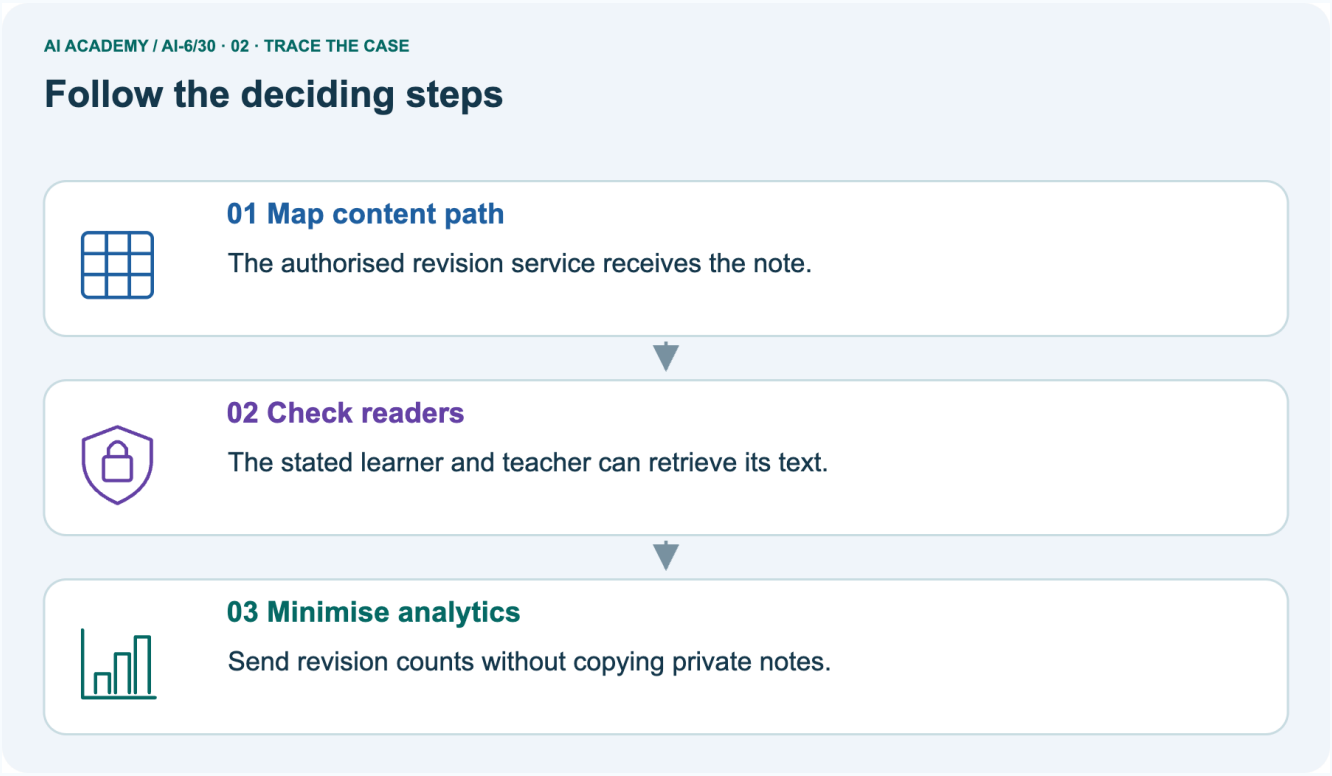
Review a private note’s path. A revision service reads a learner’s private practice note; analytics need counts only.

Only the learner and assigned teacher may read note text. The dashboard is authorised for aggregate counts, not notes.

The picture is a separate private-note path. Its policy permits note text only for the learner and assigned teacher; a broad teacher role alone is insufficient. Analytics is authorized for aggregate revision counts, not text. The added code accepts two fictional revision-kind events and emits revision_count = 2 without note text or learner IDs. A changed event containing the full note is rejected.

This is data minimization under the stated policy, not proof of anonymity: a tiny-group count may still reveal information. The dashboard’s readers, group sizes, linkability and retention remain separate review questions.

Trace authorized content and counts-only analytics



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Map content path	The authorised revision service receives the note.
2. Check readers	The stated learner and teacher can retrieve its text.
3. Minimise analytics	Send revision counts without copying private notes.

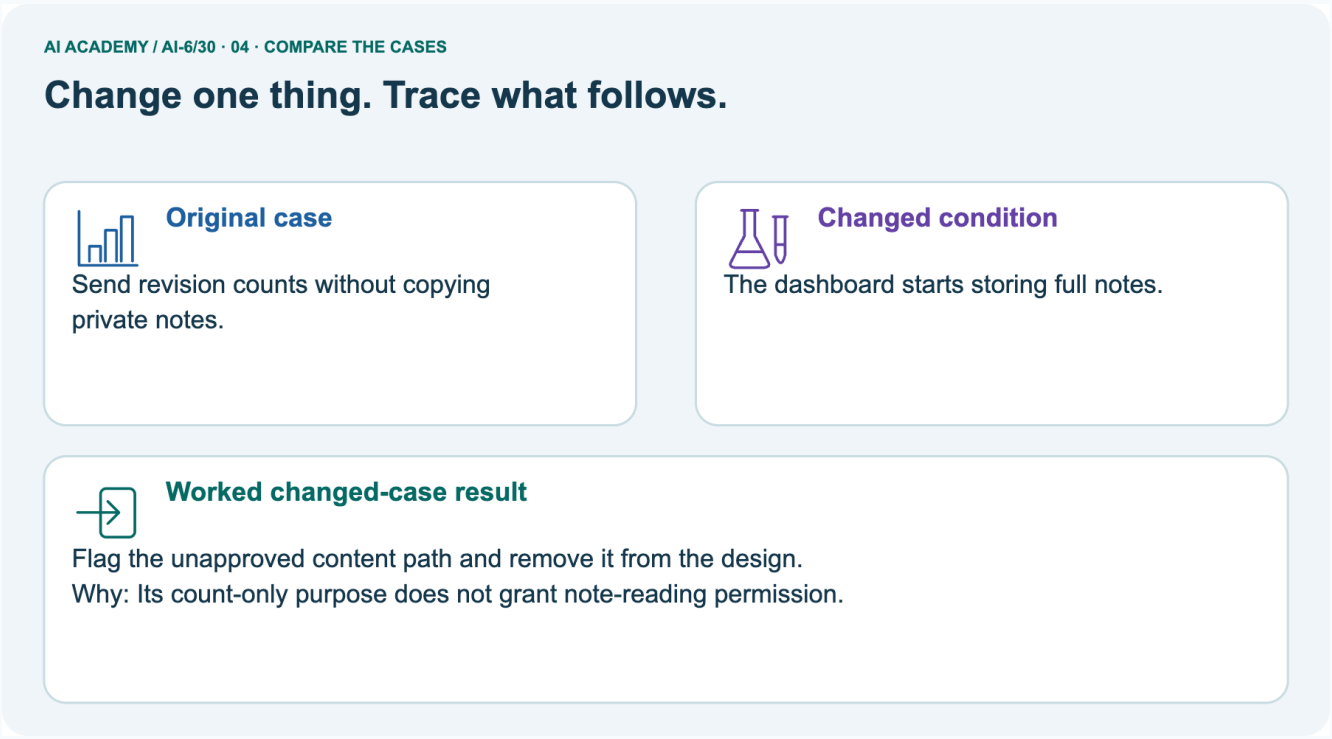
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **privacy review** follows data and authority through components. A useful dashboard does not automatically need every underlying private detail.

Predict full-note dashboard storage

- Change one thing: The dashboard starts storing full notes. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain tiny-group and linkability limits

- Tiny-group aggregates can still reveal individuals; evaluate linkability, access and retention.
- Read the labelled picture: Trace private note content and analytics counts to their permitted readers. Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?

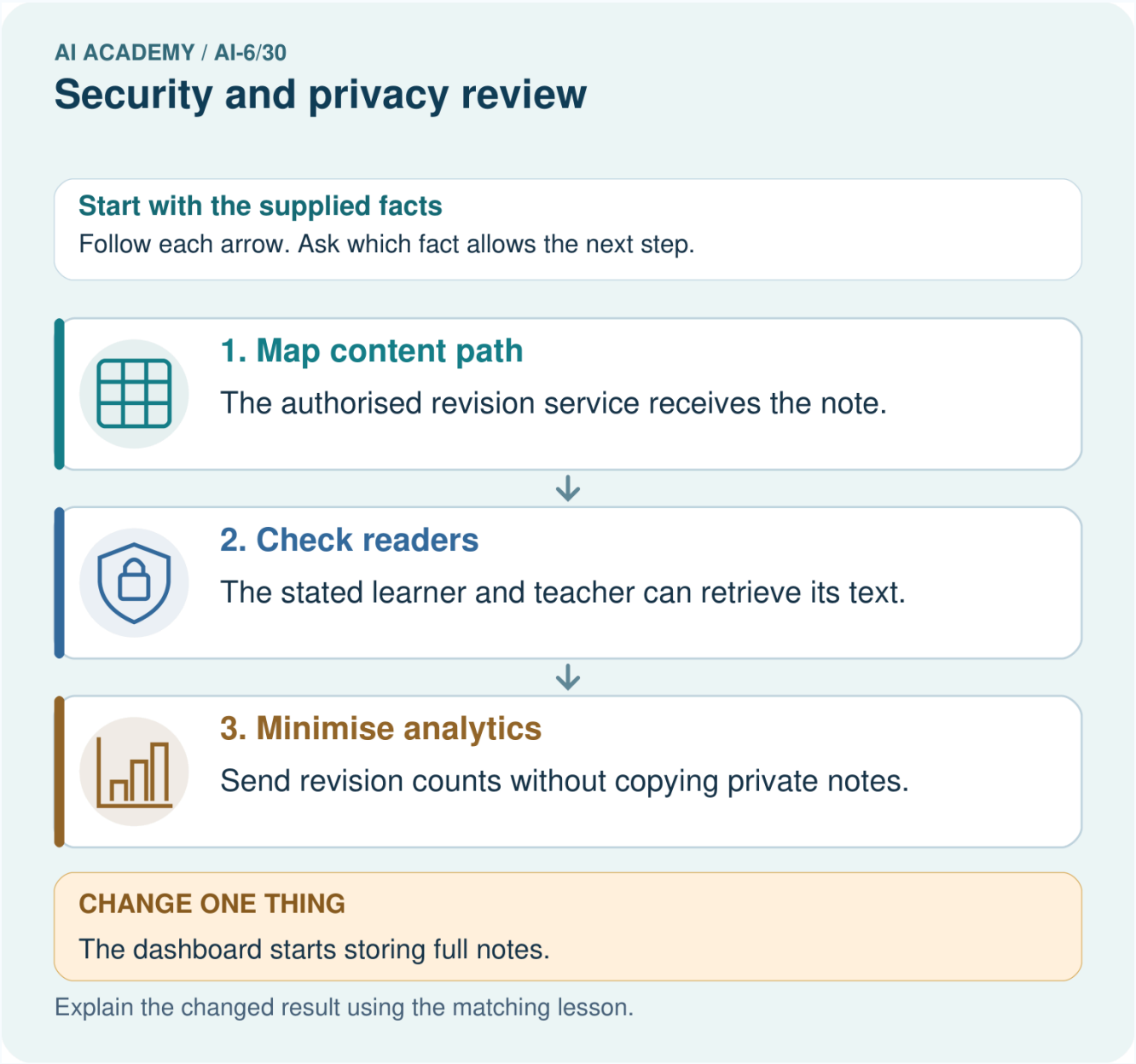
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Security and privacy review

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit copies, contexts, relationships and secondary uses

> Keep each access claim tied to its tested path and trusted policy.

Core security/privacy qualification and failure analysis

Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete misuse and minimization probes

Review probe	Expected interpretation
Direct request denied but cache returns guide	Boundary still broken; inspect both routes
Unauthorized content read then hidden in output	Already crossed context/read boundary; final redaction alone is insufficient
Request supplies role=teacher	Reject unexpected authority claim; use trusted actor fixture
Teacher not assigned to private note	Deny relationship-scoped access
Analytics event includes note text	Reject under counts-only policy
Counts omit names	Not necessarily anonymous; tiny groups/linkability still matter
Source text contains an instruction	Evidence content does not grant extra authority; never execute it as policy

Review probe	Expected interpretation
Only happy-path request tested	Misuse and denied paths remain untested
Local fixtures pass	Scoped policy evidence, not complete security or real identity verification

Guided repair and explanation

- A report omits this qualification: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise naming the asset and the bypass route

> Explain which policy check must happen before a content read.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example “Security and **privacy review**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Security and **privacy review**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Trace private note content and analytics counts to their permitted readers. Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?

> **My explanation using a pictured step and the given facts**
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Reserve the selected diagnostic-export bypass

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete declared-path and minimization audit

> Inspect denial timing, content traces and accepted analytics fields.

Current direct-denial/cache-leak investigation

Output denies teacher guide, but cache contains it under a student-accessible key. Show the intermediate steps and give the conclusion with its scope.

Runnable fictional path audit

```
"""Fictional path review; no accounts, real notes, session secrets or networks."""
from copy import deepcopy
GUIDE='FICTIONAL-RESTRICTED-GUIDE-MARKER'
TRUSTED_ROLES={'learner-1':'student','teacher-1':'teacher'}
GUIDES={'G1':GUIDE}
UNSAFE_QUERY_ONLY_CACHE={'G1':GUIDE}  # deliberately broken path, constructed for
inspection

def get_guide(actor,request,route,trace):
 if type(request) is not dict or set(request)!={'resource_id'}:return
{'status':'invalid_input'}
 rid=request['resource_id']
 if type(rid) is not str or not rid:return {'status':'invalid_input'}
 if route not in ('direct','cache'):return {'status':'invalid_route'}
 if TRUSTED_ROLES.get(actor)!='teacher':return {'status':'denied'}
 if rid not in GUIDES:return {'status':'review'}
 trace.append({'route':route,'resource_id':rid})
 # Both routes share the same trusted access gate before any content enters context.
 value=GUIDES[rid] if route=='direct' else SAFE_CACHE.get((actor,rid))
 if value is None:return {'status':'review'}
 return {'status':'allowed_fixture','content':value}
SAFE_CACHE={('teacher-1','G1'):GUIDE}

student_trace=[];teacher_trace=[]
student_results=[get_guide('learner-1',{'resource_id':'G1'},route,student_trace) for route
in ('direct','cache')]
teacher_results=[get_guide('teacher-1',{'resource_id':'G1'},route,teacher_trace) for route
in ('direct','cache')]
assert all(item=={'status':'denied'} for item in student_results) and student_trace==[]
```

```
assert all(item['content']==GUIDE for item in teacher_results) and len(teacher_trace)==2
assert GUIDE not in repr(student_results)
assert get_guide('learner-1',{'resource_id':'G1','role':'teacher'},'cache',student_trace)
['status']=='invalid_input'
assert get_guide('unknown',{'resource_id':'G1'},'direct',student_trace)
['status']=='denied'
# Separate picture: access includes relationship, not merely the broad teacher role.
NOTES={'N1':{'owner':'learner-1','assigned_teacher':'teacher-1','text':'fictional practice
note'}}
def note(actor,note_id):
 record=NOTES.get(note_id)
 if record is None:return {'status':'review'}
 if actor not in (record['owner'],record['assigned_teacher']):return {'status':'denied'}
 return {'status':'allowed_fixture','text':record['text']}
def analytics_packet(events):
 if type(events) is not list:raise ValueError('event list required')
 for event in events:
  if type(event) is not dict or set(event)!={'kind'} or event['kind']!='revision':
   raise ValueError('only minimized revision-kind events accepted')
 return {'revision_count':len(events)}
notes_before=deepcopy(NOTES)
picture_count=analytics_packet([{'kind':'revision'},{'kind':'revision'}])
assert picture_count=={'revision_count':2} and note('teacher-2','N1')=={'status':'denied'}
assert note('learner-1','N1')['text']==note('teacher-1','N1')['text']
try:analytics_packet([{'kind':'revision','text':NOTES['N1']['text']}])
except ValueError:pass
else:raise AssertionError('Full-note analytics event accepted')
assert NOTES==notes_before
print('Broken query-only cache contains the restricted marker:',GUIDE in
UNSAFE_QUERY_ONLY_CACHE.values())
print('Repaired direct/cache student decisions:',student_results,'content-read
trace:',student_trace)
print('Teacher fixture reads allowed on both routes:',[r['status'] for r in
teacher_results])
print('Separate picture minimized analytics:',picture_count,'source notes
unchanged:',NOTES==notes_before)
print('Counts can remain identifying in tiny groups. This fixture checks a declared
field/access policy, not anonymity, retention enforcement, real authentication or complete
security.')
```

Expected outcomes and untested-path limits

1. Predict both learner results and content-read traces before running or tracing. The intentionally unsafe query-only cache illustrates the stored restricted marker. Both repaired routes deny the learner before content lookup; teacher-1 may read, unexpected role fields reject and unknown actors deny.

   Separate note checks allow only its learner and assigned teacher; the count builder accepts revision-kind events and rejects note text. Source notes stay unchanged. This is not a live audit and no session secrets, real learner content or credentials are used. Source-injection, retention and unlisted routes still need their own tests; the code does not claim to cover them. Mark paper traces “not run”.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: inspect a diagnostic export after direct denial

> A new return path needs the same confidentiality boundary.

Complete selected independent task

Service blocks direct teacher lookup but diagnostic export includes full teacher answers. Identify defect and a test.

First attempt and support record

> **My independent explanation, trace and limits**
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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply a new case independently

Attempt this without hints

Service blocks direct teacher lookup but diagnostic export includes full teacher answers. Identify defect and a test.

> **My independent procedure, result and limitation**
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve
</details>

MAIN LESSON · SECTION 11

Save an asset/actor/path map with denied-case evidence

> Record only permitted metadata and fictional test markers.

Project task and artifact

Add a labelled security and **privacy review evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of security and **privacy review**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local run or paper trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the tested paths and the review gaps

> A clean output cannot excuse restricted content entering an unauthorized context.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of security and **privacy review**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Threat model	Assets, actors and failure assumptions considered
Attack surface	The inputs and routes through which behaviour can be influenced
Privacy review	A check of purpose, access and data handling

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Security and privacy review — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Output denies teacher guide, but cache contains it under a student-accessible key.	For the classroom service identify restricted guides, session secrets and student content. Test denied roles, unexpected fields, source injection and minimised logging without real credentials.	Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student output/context.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Review a private note's path. A revision service reads a learner's private practice note; analytics need counts only.

By the end, I can	How I will show it
Explain the mechanism and terminology of security and privacy review	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Review a private note’s path. A revision service reads a learner’s private practice note; analytics need counts only.
- Which labelled step matches this detail? Send revision counts without copying private notes.
- 1. Map content path
- 2. Check readers
- 3. Minimise analytics
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Minimise analytics Send revision counts without copying private notes. A **privacy review** follows data and authority through components.

A useful dashboard does not automatically need every underlying private detail.

Changed-case question and optional hint

- Changed-case practice: The dashboard starts storing full notes. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Flag the unapproved content path and remove it from the design.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: Flag the unapproved content path and remove it from the design. Its count-only purpose does not grant note-reading permission.

Repair and retain the record

Review Performance and cost experiments (ai-6/29). Goal: Explain the **mechanism** and terminology of performance and cost experiments. Short practice: Calculate per-case seconds and cost units for A and B.

Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?. Return to this week and explain what you changed.

Review Human escalation workflows (ai-6/27). Goal: Explain the **mechanism** and terminology of human escalation workflows. Short practice: Identify the exact service boundary in the bottle question.

What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?. Return to this week and explain what you changed.

Review Release and rollback (ai-6/24). Goal: Explain the **mechanism** and terminology of release and rollback. Short practice: Compute A's and B's correctness percentages.

Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

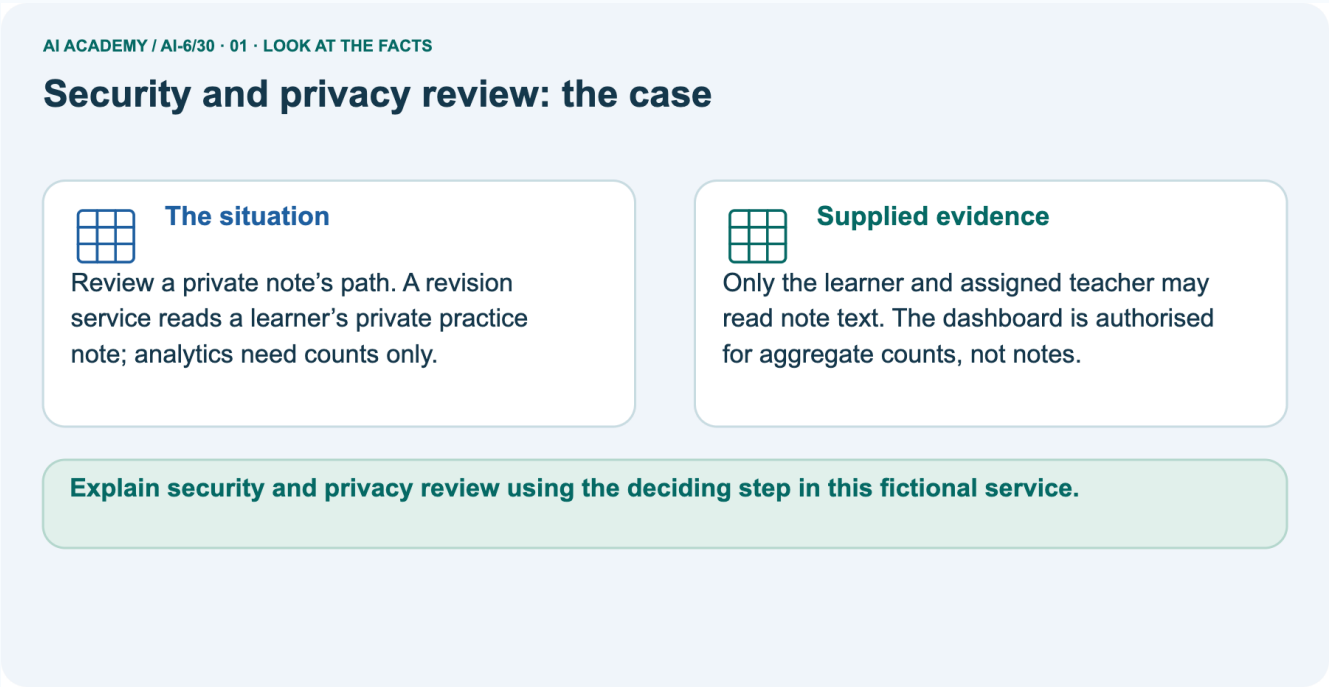
Keep your completed work

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Security and privacy review: the case



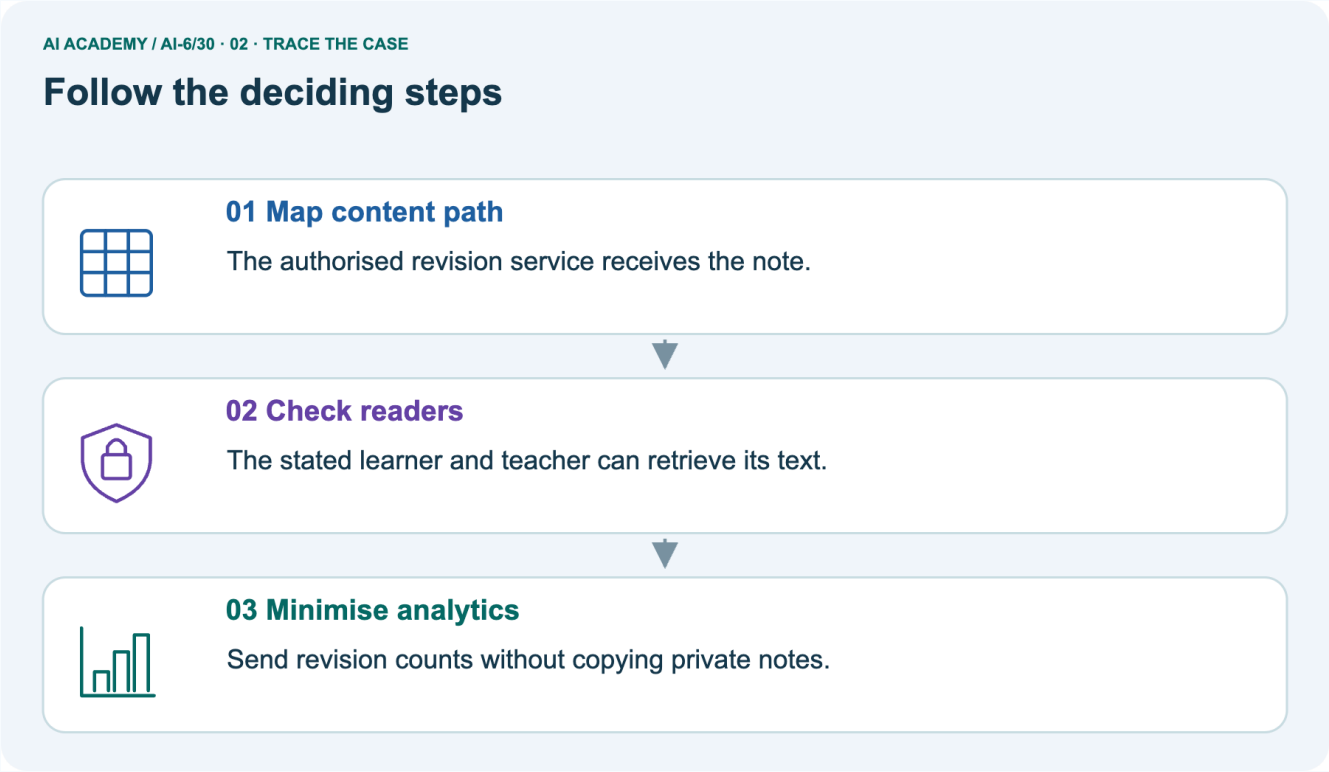
Follow the labelled picture with the matching explanation. Open the vector diagram

Review a private note's path. A revision service reads a learner's private practice note; analytics need counts only. Supplied facts: Only the learner and assigned teacher may read note text.

The dashboard is authorised for aggregate counts, not notes.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

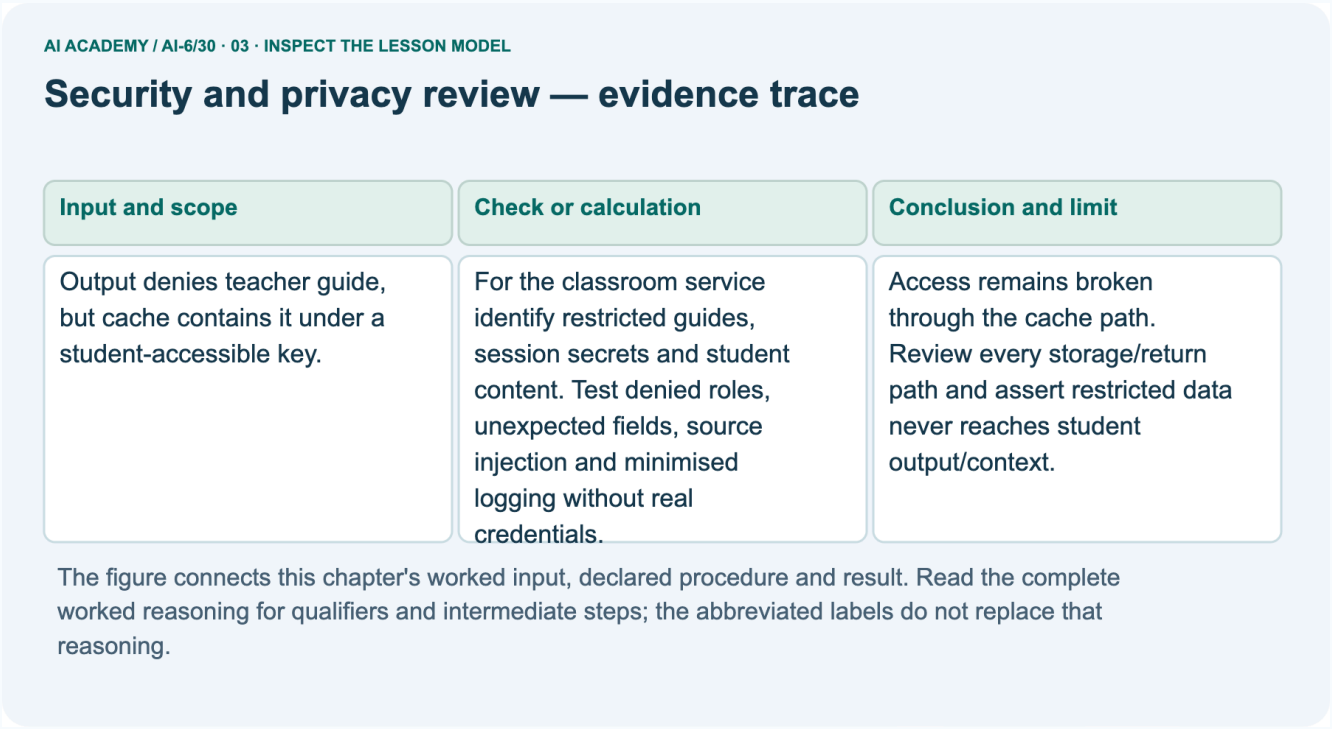


Follow the labelled picture with the matching explanation. Open the vector diagram

Map content path: The authorised revision service receives the note. → Check readers: The stated learner and teacher can retrieve its text. → Minimise analytics: Send revision counts without copying private notes.

Point to each numbered stage and explain which recorded **input** it uses.

03. Security and privacy review — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

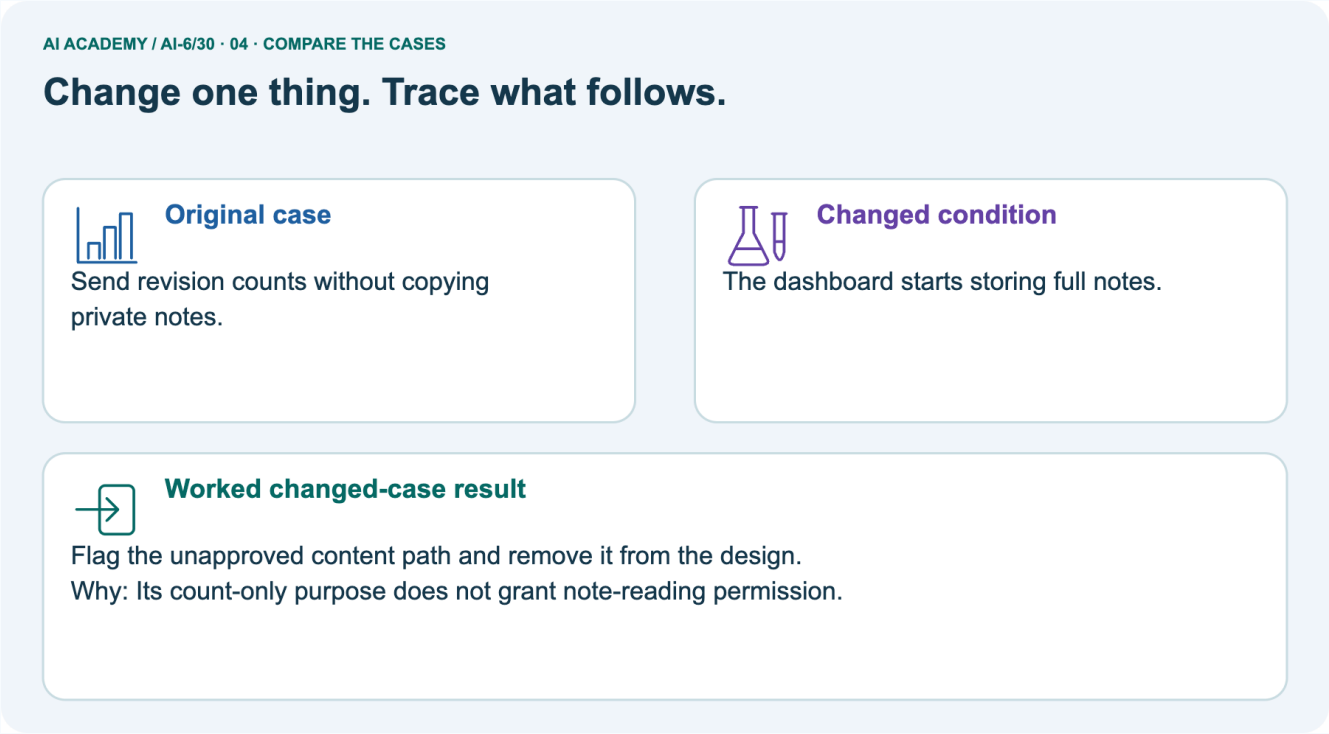
Security and **privacy review** — **evidence** trace. **Output** denies teacher guide, but cache contains it under a student-accessible key. | For the classroom service identify restricted guides, session secrets and student content.

Test denied roles, unexpected fields, source injection and minimised logging without real credentials. | Access remains broken through the cache path.

Review every storage/return path and assert restricted data never reaches student **output**/context.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



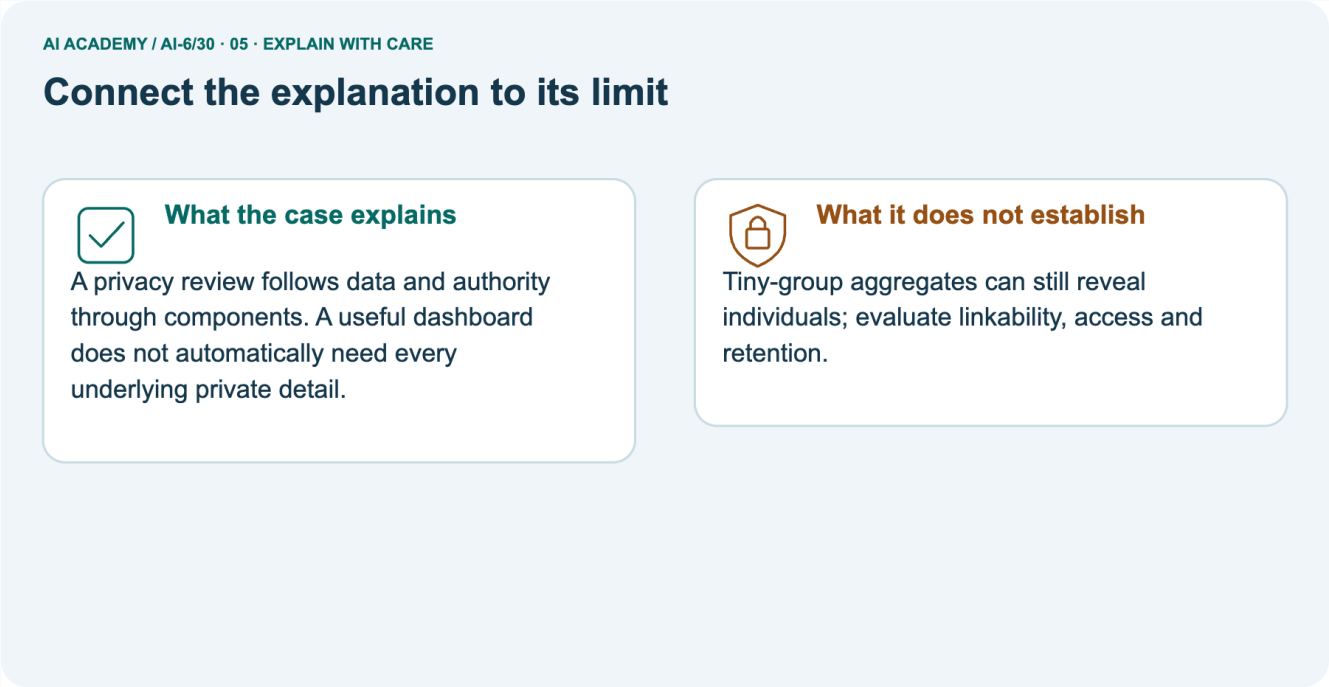
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Send revision counts without copying private notes. Changed condition: The dashboard starts storing full notes. Expected result: Flag the unapproved content path and remove it from the design.

Reason: Its count-only purpose does not grant note-reading permission.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



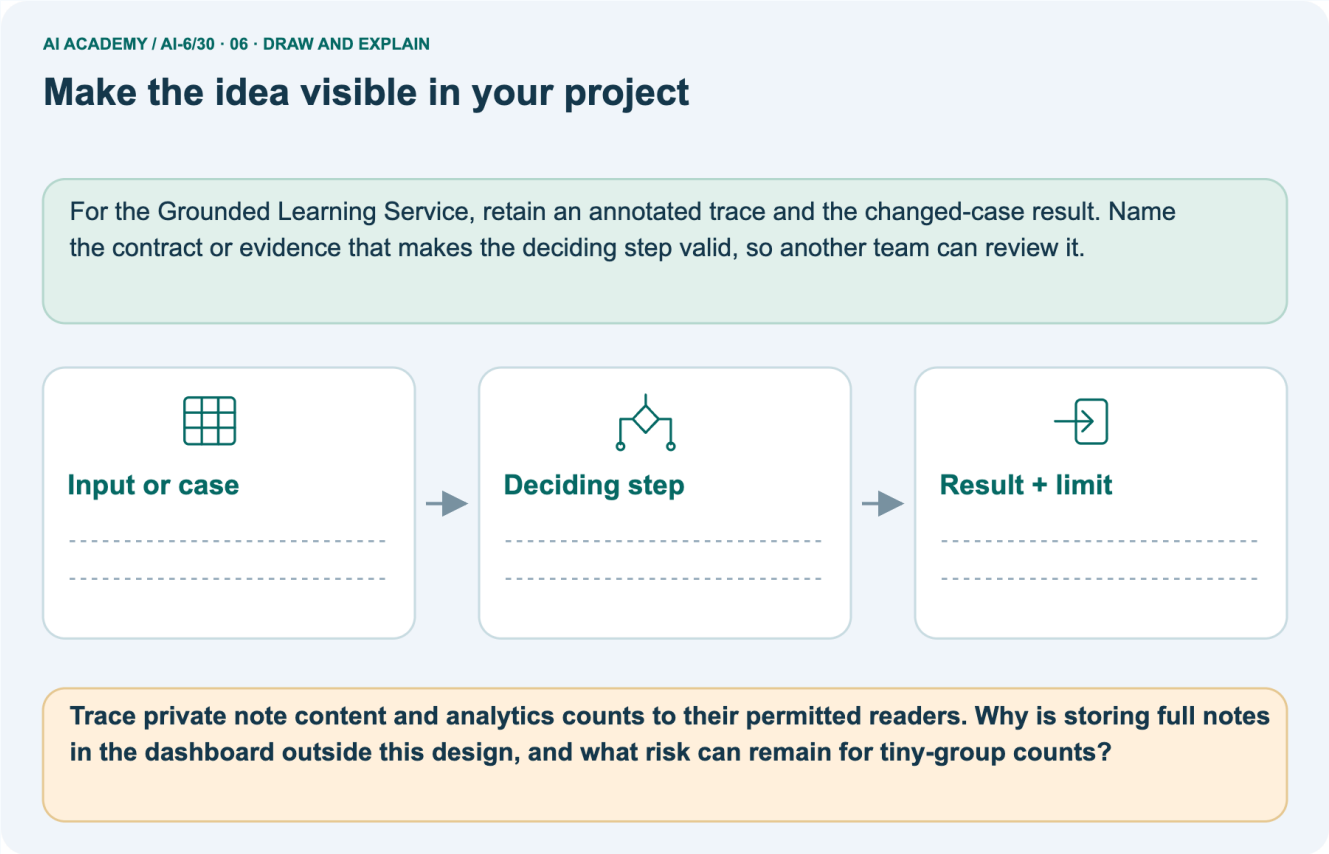
Follow the labelled picture with the matching explanation. Open the vector diagram

A **privacy review** follows data and authority through components. A useful dashboard does not automatically need every underlying private detail.

Limit: Tiny-group aggregates can still reveal individuals; evaluate linkability, access and retention.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Security and privacy review: the case

02. Follow the deciding steps

03. Security and privacy review — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.